Original research

International Journal of Sports Science & Coaching

International Journal of Sports Science & Coaching
2023, Vol. 18(2) 446–452

DOI: 10.1177/17479541221079531
journals.sagepub.com/home/spo
SAGE

Effects of repeated-sprint training in hypoxia induced by voluntary hypoventilation on performance during ice hockey off-season

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Abstract

This study aimed to assess the effects of an off-season period of repeated-sprint training in hypoxia induced by voluntary hypoventilation at low lung volume (RSH-VHL) on off-ice repeated-sprint ability (RSA) in ice hockey players. Thirty-five high-level youth ice hockey players completed 10 sessions of running repeated sprints over a 5-week period, either with RSH-VHL ($n = 16$) or with unrestricted breathing (RSN, $n = 19$). Before (Pre) and after (Post) the training period, subjects performed two 40-m single sprints (to obtain the reference velocity (V_{ref})) followed by a running RSA test (12×40 m all-out sprints with departure every 30 s). From Pre to Post, there was no change in V_{ref} or in the maximal velocity reached in the RSA test in both groups. In RSH-VHL, the mean velocity of the RSA test was higher (88.9 ± 5.4 *vs.* 92.9 ± 3.2% of V_{ref}; $p < 0.01$) and the percentage decrement score lower (11.1 ± 5.2 *vs.* 7.1 ± 3.3; $p < 0.01$) at Post than at Pre whereas no significant change occurred in the RSN group (89.6 ± 3.3 *vs.* 91.3 ± 1.9% of V_{ref}, $p = 0.11$; 10.4 ± 3.2 *vs.* 8.7 ± 2.3%; $p = 0.13$). In conclusion, five weeks of off-ice RSH-VHL intervention led to a significant 4% improvement in off-ice RSA performance. Based on previous findings showing larger effects after shorter intervention time, the dose-response dependent effect of this innovative approach remains to be investigated.

Keywords

Altitude training, fatigue, repeated-sprint ability, team sport

Introduction

Among team sports, ice hockey is inimitable due to its highly complex technical skills and physical requirements.[1–3] Contemporary ice hockey time-motion analysis shows that fatigue develops as the game progresses.[4] In this view, conditioning programs focusing on neuromuscular fatigue resistance, notably through repeated-sprint ability (RSA) training, have gained in popularity.[5,6] Block periodization[7] during off- (~10–12 weeks) or pre-season (~5–7 weeks) has been shown to provide putative physical/physiological gains in male ice hockey players.[8–10]

In recent times, it has been reported that performing a repeated-sprint training with oxygen deprivation (the so-called repeated-sprint training in hypoxia, RSH) was very effective to improve the RSA-related limiting factors (*e.g.* muscle excitability and recruitment, neural drive, metabolic energy supply and by-products).[11,12] Despite the increased availability of hypoxic devices (hypoxicators, altitude tents and chambers),[13] training in hypoxia is not accessible to all athletes (*e.g.* specific logistic, cost). As an alternative, the use of voluntary hypoventilation at low lung volume (VHL), which consists in performing short bouts of end-expiratory breath holding while

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exercising, could induce severe drop (down to ~87%) in arterial oxygen saturation (SpO_2)[14,15] leading to tissue hypoxia.[16,17] Although the hypoxic stimulus is not continuous when using this method, so that the physiological adaptations may not be similar to those of training in systemic hypoxia, exercise with VHL provides an additional physiological stress to produce greater or more rapid adaptations and consecutive performance enhancement compared to similar training in normoxia (*i.e.* with unrestricted breathing). When combined with repeated sprints, the RSH-VHL approach has proved putative benefits in swimming,[18] cycling,[19] rugby[20] and basket-ball.[21] The mechanisms induced by RSH-VHL include higher anaerobic tolerance[18,19] and oxygen uptake,[19] two major determinants of RSA performance.[6] While it has been shown that holding breath at high lung volume during exercise could also induce severe arterial desaturation[22] and may help reduce blood acidosis and oxidative stress,[23] only the VHL technique can provoke a sufficiently fast drop in SpO_2 to perform a RSH effort.[18–21] Due to the specific features of ice hockey (*e.g.* frequent shifts inducing short high-intensity sprints, partial occlusion in lower limbs occurring while skating, large neuromuscular fatigue), one may hypothesize that RSH-VHL would be particularly appropriate in ice hockey. Since, to our knowledge, there is no hypoxic ice rink, the VHL method seems to be a relevant alternative for on-ice training.

So far, there is little interest shown on the off-season period in ice hockey. This period generally includes an active off-ice phase where non-specific exercises are performed. The use of off-ice exercises, such as run-based training, during the off-season have been endorsed to improve on-ice skating performance.[10] Interestingly, the physiological adaptations induced by RSH-VHL are deemed to lead to transferable performance benefits from one exercise mode to another (*e.g.* from cycling- to running-based performance).[24] Using RSH-VHL may therefore provide a relevant perspective for improving the off-ice phase of the off-season period in ice hockey players. Furthermore, to date, all of the RSH-VHL studies have been conducted over two to four weeks of training. The choice of such short periods of time probably pertains to the demanding nature of the RSH-VHL approach. While this method seems to induce fast performance gains, it may also lead to fatigue after several weeks of application.[19,20] It would however be interesting to assess whether a (slightly) longer training period could lead to greater benefits.

Within this context, the original purpose of this study was to assess the effects of a 5-week off-ice RSH-VHL on off-ice RSA performance in high-level youth ice hockey players. We hypothesized that RSH-VHL would induce a larger off-ice physical enhancement compared to similar training performed with unrestricted breathing.

Table 1. Characteristics of ice-hockey players who completed the study.

	RSN	RSH-VHL
U17 Team (n)	7	5
U20 Team (n)	12	11
Age (y)	16.5 ± 1.5	16.9 ± 1.4
Height (cm)	175 ± 8.5	174 ± 8.3
Weight (kg)	73.6 ± 15	71.2 ± 13

RSN, repeated-sprint training with unrestricted breathing. RSH-VHL, repeated-sprint training in hypoxia induced by voluntary hypoventilation at low lung volume; Data for age, height and weight are mean ± SD.

Methods

Subjects

Forty-three high-level youth male ice hockey players were recruited to participate in this study among which thirty-five could complete the entire protocol. Eight out of the 43 subjects were unable to complete the protocol due to injuries (n = 3), unavailability (n = 4) or club transfer (n = 1) and were therefore excluded from the data analysis. The subjects were members of two junior teams (U17 and U20) belonging to the same ice hockey club and reported at least eight years of ice hockey practice. The U20 players were engaged in the Swiss ice hockey federation's highest level U20 national championship (U20-Elite). The U17 participants took part in the Swiss ice hockey federation's second highest level's U17 national championship (U17-Top). The main descriptive characteristics of the subjects who completed the entire protocol are presented in Table 1. All subjects were lowlanders (*i.e.* < 1000 m), were not acclimatized or recently exposed to altitude/hypoxia before or during the experiment and had never experienced any hypoxic or VHL training methods. The study was approved by the Ethical Commission for Human Research (Lausanne, Switzerland) and complied with the Declaration of Helsinki. All subjects and their parents/legal guardians (for minors), were informed about the risks, benefits, and procedures of the study before providing their written informed consent prior to participation.

Procedures

The experiment was conducted during the off-season period (from mid-April to mid-June), where the subjects did not have access to on-ice facilities. Following full regeneration from the previous season (unsupervised light recreational physical activity only), subjects restarted their off-ice training through 5-6 sessions per week (for a total of 7–9 h) of strength and conditioning including strength, speed, agility, plyometric and aerobic-based running exercises. For the purpose of the study, considering that the subjects could not use sport-specific exercises, two weekly aerobic running sessions were replaced by running repeated-sprint sessions for five consecutive weeks. A running RSA test was implemented one week before (Pre) and one week after (Post) the training period. The subjects were matched into pairs according to their team age group (U17 or U20) and RSA performance at Pre, and randomly assigned to either RSH-VHL (n = 21) or the group who performed similar training with unrestricted breathing (RSN, n = 22). As described elsewhere,[17,19,20] the RSH-VHL method consists in performing a first exhalation down to the functional residual capacity and then to maximally sprint while holding the breath. Just after the release of the breath holding, a second exhalation down to the residual volume must be done in order to evacuate the carbon dioxide accumulated in the lungs. In order to allow an appropriate learning of the breathing technique, two familiarization sessions were scheduled. The VHL approach has been recognized as a hypoxic training method potentially useful for a wide range of sports[12,25] and has been recently included in the updated nomenclature of altitude/hypoxic training methods.[26,27]

Training protocol. Within the off-season training period, ten repeated-sprint sessions were implemented over five weeks (two sessions per week 48 h apart) on an outdoor soccer field. For practical reasons, training sessions were scheduled at the same time of day (± 1 h) on Tuesday and Thursday for the U20 group and on Wednesday and Friday for the U17 group. Each session included a 10-min standardized warm-up (active mobilisation, dynamic stretching, running drills and progressive accelerations) and running repeated 40-m "all-out" sprints (with departure every 30 s). As described elsewhere,[20] similar work-to-rest ratio (*i.e.* 1:4) was selected with a sprinting distance of 40 m corresponding to the time (∼ 6 s) that permits to maintain end-expiratory breath holding. For the first session, subjects performed two sets of 6 repetitions. The number of repetitions was then progressively increased over the subsequent sessions according to each individual rating of perceived exertion (RPE). If the total number of repetitions exceeded 16, a third set including at least 6 repetitions was implemented. The maximum number of repetitions per set (*i.e.* 8) as well as the training load for the repeated-sprint sessions were determined based on previous RSH-VHL studies.[18–20] In all cases, each set was separated by 3 min of semi-active recovery (*i.e.* walking). The RSH-VHL group completed the sprints with VHL and the RSN group with unrestricted breathing. In both groups, the recovery between sprints and sets was performed with unrestricted breathing.

Testing protocol. Pre- and Post-testing sessions consisted of a running RSA test performed on an indoor athletics track and including twelve straight-line 40-m "all-out" sprints (with unrestricted breathing) departing every 30 s. Prior to testing, the same warm-up as during the training sessions (see above) was completed. It was followed by two single "all-out" 40-m running sprints (5 min apart) to determine the reference velocity (V_{ref}) which was calculated from the best performance. Thereafter, the RSA test was performed in alternating directions with subject semi-actively recovering (*i.e.* walking back to the finishing/starting line). A countdown was given 5 s before the start in order to allow the subjects to assume the standing ready position (with shifted feet behind the starting line).

Strong verbal encouragements were given to all subjects during each sprint. To avoid any pacing strategies, the subjects were asked to reach at least 90% of V_{ref} in the first repetition. If they failed to do so, they had to start the test again after a resting period of 5 min. Pre-testing schedule was reproduced at Post to allow the same day and time for each subject. Within the 24 h preceding the testing sessions, subjects were instructed to avoid high-intensity exercise and to refrain from consuming caffeine and alcohol.

Measurements

Training data. To evaluate the total training load (including all off-ice training), all subjects were asked to report both the total duration and the session RPE of each training during the 5-week period. Training load was calculated using the method developed by Foster et al.[28] which consists of multiplying the session RPE by its duration which has been validated in ice hockey.[29] For each subject, SpO_2 and heart rate (HR) were recorded during a single training session (four subjects per training session) between the third and fifth week of the training intervention. SpO_2 was continuously monitored with a portable wrist pulse oximeter (Nonin, Playmouth, USA). The lowest SpO_2 values recorded after the completion of each sprint (generally after 6 to 10 s of recovery) were used to assess the minimal and average SpO_2 over an entire set. HR was also continuously measured (Polar Oy, Kempele, Finland) to obtain the maximal (HR_{max}) and the mean (HR_{mean}) HR values over an entire set (including the recovery periods following each repetition).

Testing data. The RSA performance was measured using photocells connected to an electronic timer (Witty, Microgate, Bolzano, Italy). Maximal (V_{max}) and mean velocity (V_{mean}) were calculated from the fastest repetition (generally the first one) and the twelve 40-m sprinting times, respectively. Both V_{max} and V_{mean} were expressed as percentage of V_{ref} (best performance of the two single "all-out" 40-m sprints).[20] HR was continuously monitored throughout the RSA test (Polar Oy, Kempele, Finland), with HR_{mean} from the first to the twelfth repetition (including recovery periods) considered for subsequent analysis. Perception of effort for the RSA test was assessed immediately after the last sprint using session RPE (0-10).[29] The RSA fatigue was evaluated by calculating the percentage decrement score (S_{dec}) as follows:

$$(\text{total sprint time} \ / \ \text{ideal sprint time}) - 1] \times 100)$$

where total sprint time is the sum of the 12 sprinting times and ideal sprint time is the sum of the theoretical 12 fastest sprinting times. This formula has been found to be the most valid and reliable method to estimate RSA-induced fatigability.[30]

Statistics. Two-way ANOVA with repeated measures were used to determine whether there was an effect of training intervention (RSN *vs.* RSH-VHL) and/or time (Pre *vs.* Post) in the mean of the variables measured during the RSA test. We also used two way repeated-measures ANOVA to analyse the change in velocity throughout the 12 sprints of the RSA test within each group. When an effect of time and/or training intervention was found, the Bonferroni *post hoc* procedure was used to localize the difference.[31] Training data were compared between groups with student *t*-test All data analyses were conducted with the software Sigmastat 3.5 (Systat Software, CA, USA). Data are presented as mean ± SD. Significance level was set at $p < 0.05$.

Results

Training data

Among the 35 subjects who completed the entire protocol, there was no difference between the RSH-VHL (n = 16) and the RSN groups (n = 19) in the total training load over the 5-week period (21177 ± 3772 *vs.* 19912 ± 3494 arbitrary units, $p = 0.31$). There was also no difference between groups in the total number of 40-m sprints (167 ± 19 *vs.* 165 ± 22, $p = 0.79$), the mean number of repetitions per session (17.5 ± 1.2 *vs.* 17.7 ± 1.3, $p = 0.75$) and the mean RPE per session (8.1 ± 0.5 *vs.* 8.0 ± 0.5, $p = 0.49$).

During a single training session, HR_{max} (184 ± 7 bpm *vs.* 184 ± 8 bpm, $p = 0.61$) and HR_{mean} (174 ± 8 bpm *vs.* 175 ± 9 bpm, $p = 0.71$) were not different between RSH-VHL and RSN whereas both the minimal SpO_2 (82.5 ± 4% *vs.* 93.5 ± 1.5%, p < 0.01) and average SpO_2 (88.8 ± 2.4% *vs.* 95.8 ± 1.0%; p < 0.01) were lower in RSH-VHL (p < 0.01).

Testing data

While there was no difference between groups in all variables both at Pre and Post, V_{mean} was higher and S_{dec} lower after the training period in RSH-VHL (p < 0.01) whereas they remained unchanged in RSN (p = 0.11 and 0.13, respectively) (Table 2). Both V_{ref} and V_{max} reached in the RSA test were not different between Pre and Post in both groups. On the other hand, there was a significant decrease in HR_{mean} from Pre to Post both in RSH-VHL and RSN. The RPE was higher at Post in RSN only. The analysis of each of the twelve 40-m sprints throughout the RSA test shows that at Post compared to Pre, the relative velocity was lower in RSN at the 2nd sprint and higher both in RSH-VHL and RSN from the 7th sprint to the end of the test (Figure 1).

Discussion

This study investigated the effects of 10 running RSH-VHL sessions over a five-week period on off-ice RSA performance in high-level youth ice hockey players in the particular context of the off-season training period. The results indicate greater improvements in RSA performance (higher V_{mean}, lower S_{dec}) in RSH-VHL after the training period whereas no significant change occurred in the group which performed similar training with unrestricted breathing (RSN). These findings report RSH-VHL as a potential alternative conditioning method in ice hockey.

Several studies have already demonstrated the effectiveness of RSH-VHL for improving RSA in sport-specific

Table 2. Data of the single "all-out" 40-m sprints and the RSA test before (Pre) and after (post) RSH-VHL and RSN.

	RSH-VHL		RSN		ANOVA *P* Values		
	Pre	Post	Pre	Post	T	C	T × C
V_{ref} ($m.s^{-1}$)	6.95 ± 0.4	6.95 ± 0.4	6.80 ± 0.3	6.89 ± 0.4	0.55	0.32	0.52
V_{mean} (%V_{ref})	88.9 ± 5.4	**92.9 ± 3.2****	89.6 ± 3.3	91.3 ± 1.9	< 0.01	0.53	0.12
V_{max} (%V_{ref})	98.3 ± 6.2	99.3 ± 3.3	99.4 ± 3.4	98.2 ± 2.1	0.32	0.37	0.40
S_{dec} (%)	11.1 ± 5.2	**7.1 ± 3.3****	10.4 ± 3.2	8.7 ± 2.3	< 0.01	0.55	0.10
HR_{mean} (bpm)	184 ± 6.0	**180 ± 6.5***	185 ± 7.2	**180 ± 10.9***	< 0.01	0.79	0.78
RPE	8.8 ± 1.2	9.2 ± 0.6	8.4 ± 1.0	**9.1 ± 0.6***	< 0.01	0.43	0.35

Values are mean ± SD. RSH-VHL, repeated-sprint training in hypoxia induced by hypoventilation at low lung volume; RSN, repeated-sprint training with unrestricted breathing; V_{ref}, reference velocity obtained from the best of the two single "all-out" 40-m sprints; V_{mean}, mean velocity expressed as % of V_{ref}; V_{max}, maximal velocity expressed as % of V_{ref}; S_{dec}, percentage decrement score; HR_{mean}, mean heart rate; RPE, rating of perceived exertion; T, time effect; C, condition effect; I, interaction effect (time × condition). Values in bold show significant difference, *$p < 0.05$ for difference with Pre; ** $p < 0.001$ for difference with Pre.

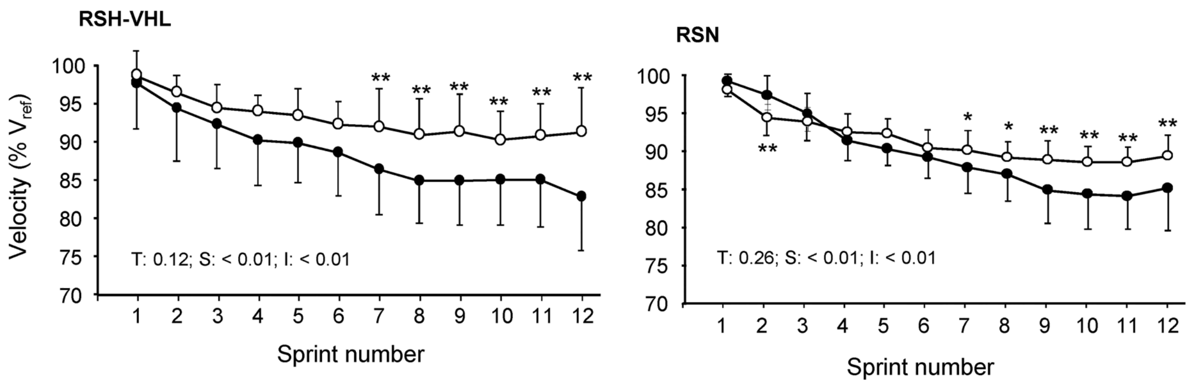

Figure 1. Velocity expressed as a percentage of the reference velocity (v_{ref}) for each repetition (n = 12) of the running repeated 40-m "all-out" sprint test (departing every 30 s) performed before (black circles) and after (white circles) repeated-sprint training in hypoxia induced by voluntary hypoventilation at low lung volume (RSH-VHL) or with unrestricted breathing (RSN). T: Time effect (Pre *vs.* Post); S: Sprint number effect; I: Interaction effect (time x condition). * $p < 0.05$, ** $p < 0.01$, for significant difference with the same sprint at Pre.

exercises. In open-loop RSA test designs (*i.e.* maximum number of sprints performed until task failure), highly-trained swimmers[18] and rugby players[20] markedly increased RSA performance after two weeks (6 sessions) and four weeks (7 sessions) of RSH-VHL, respectively, while no change was observed in the control groups who performed similar exercise with unrestricted breathing. In a close-loop RSA test design (*i.e.* with a predefined number of sprints), the mean power output was increased by ~8% in cyclists after RSH-VHL (6 sessions within 3 weeks) compared to control (+2%).[19] Likewise, basket-ball players of national level who trained for four weeks with RSH-VHL also reduced fatigue to a greater extent than the control group.[21] The 4% increase in V_{mean} reported in the present study after five weeks of RSH-VHL is difficult to compare to previous studies using open-loop RSA tests. On the other hand, it is noteworthy that this increase is twofold lower than the aforementioned gains reported after only 3 weeks of cycling RSH-VHL.[19] Considering that in this latter study subjects were also highly-trained and performed similar work-to-rest ratio (1:4; 6-s sprinting and 24-s recovery) for an equal number of sprints per session (17.8 *vs.* 17.5), our results tend to demonstrate that, when using RSH-VHL, a longer period of training may not induce the greatest gains in RSA performance. While the RSH-VHL dose-response effect may require additional investigation, it seems that an addition of only 1–2 weeks of RSH-VHL may be physically and psychologically challenging. This method should therefore be used carefully to avoid nonfunctional overreaching or overtraining that could lead to counterproductive effects (*i.e.* unchanged or even decreased performance).[32]

It is interesting to note that during the RSA test, the increase in velocity from Pre to Post in RSH-VHL was observable from the fourth 40-m sprint (Figure 1), with a greater (significant) difference from the seventh sprint. Such changes have also been observed during the second part of a RSA test following RSH-VHL[19,21] or high-intensity VHL training.[24] Although not measured in the present study, it is tempting to associate the putative benefit of RSH-VHL with the increase in oxygen uptake previously reported in a cycling-based RSH-VHL study.[19] These authors attributed the augmented oxygen uptake to a greater stroke volume which may be the consequence of the large ventricular filling and the "pump effect" that occur at the end or just after each bout of end-expiratory breath holding.[16,17,33] This central adaptation assumption was recently reinforced by the larger muscle re-oxygenation observed during the recovery periods of a RSA test without concomitant greater oxygen extraction following four weeks of RSH-VHL in basketball players.[21] Considering that in the present study the SpO_2 levels recorded during training in RSH-VHL were similar to those of the previous studies, it is likely that the physiological adaptations were also similar. While an increase in oxygen uptake may have participated in energy supply during the 40-m sprints, it probably played the major role during the recovery periods following sprints. Indeed, a higher oxygen availability during the recovery periods of repeated-sprint exercise allows to increase the resynthesis of phosphocreatine, which is the main energy component in repeated sprints,[6,34] as well as the removal of waste metabolites (*i.e.* inorganic phosphate, hydrogen ions). Moreover, the influence of the oxygen uptake kinetics on RSA is well established.[35]

From a practical point of view, the present results indicate that ice hockey players may potentially optimize the off-ice period while using RSH-VHL during the off-season. It has been recently shown that the performance benefits after high-intensity VHL training could be transferable from one exercise modality to another,[24] probably due to the central adaptations mentioned above. Furthermore, a recent study has just demonstrated the interest of off-ice resisted sprints for on-ice skating.[36] Therefore, we can confidently speculate on the putative benefits of a running

RSH-VHL on ice hockey performance. The present study also confirms the usefulness of this new alternative to altitude/hypoxic training for practitioners and players who do not have access to natural altitude or simulated hypoxic facilities. Altogether, by improving ice hockey-specific physical requirements such as RSA performance and resistance to fatigue, the RSH-VHL approach may provide a decisive advantage in the final outcome of an ice hockey game. It can for instance give the possibility to win possession of the puck and increase the chances of scoring while preventing the opponents to do so.[4]

As for all VHL studies, the main limitation of the present one is that blinding was not possible. Therefore, a placebo effect cannot be excluded in RSH-VHL whereas a nocebo effect may have occurred in RSN, possibly illustrated by the likely pacing effect during Post-test. Interestingly, in their latest VHL study, Woorons et al.[24] used a protocol in which the subjects of the control group trained in a normoxic room which they were told was placed under hypoxic conditions. However, they did not improve performance whereas the subjects of the VHL group did. This result, associated to the physiological adaptations previously reported after RSH-VHL[18,19,21] illustrate a negligible impact of placebo/nocebo effect. Other uncontrolled effects such as different national standard and age-dependent fitness profiles,[37,38] possibly related to different muscle mass, strength and power, all critical parameters of RSA performance,[39] could not be excluded.

Conclusion

High-level youth ice hockey players who performed five weeks of off-ice repeated-sprint training in hypoxia induced by voluntary hypoventilation at low lung volume improved RSA performance (*i.e.* mean velocity and resistance to fatigue) to a larger extent than similar training with unrestricted breathing. Such an approach may therefore represent a successful alternative to altitude/hypoxic training. However, based on previous findings showing larger effects after shorter intervention time, the dose-response dependent effect of this innovative approach remains to be investigated.

Acknowledgements

The authors would like to sincerely thank all the ice hockey players who participated in this study and gave their best all along the testing and training sessions. The authors also express their gratitude to the coaching staffs of both U17 and U20 squads, the director of the club and all the people who helped in the implementation of the study and made it possible.

The authors declare that they have no conflict of interest and have not received any funding for this work.

All underlying research materials related to this paper can be accessed by contacting directly Dr Franck Brocherie or Dr Xavier Woorons.

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